

Tab 1

**Forecasting Cyber Attack:
Evaluating a Darwin-Gödel Cyber Range for Preemptive Cyber Defense**

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Cybersecurity decision-makers at leading global organizations, commonly known as Chief Information Security Officers (CISOs), are increasingly tasked with guiding AI-driven transformation initiatives to stay ahead of adversaries in 2026. Cyber threats now include nation-state and criminal actors who are developing their own AI models and tools to rapidly identify vulnerabilities, exploit weaknesses, and monetize organizations with outdated or inadequate security postures. In response to such AI-enabled offensive capabilities, nations are increasingly emphasizing AI-driven defense, recognizing that acting first may be more critical than acting fast. For instance, Japan recently enacted the Active Cyber Defense Law, granting its Defense Forces the authority to preemptively engage cyber targets when deemed sufficiently threatened (Dobell & O'Neill, 2025). Yet, how elite AI and human defense mechanisms can reliably learn to anticipate and preempt attacks remains an open question.

In 2025, the first documented large-scale AI-driven cyber attack successfully targeted global firms entirely without human intervention (Anthropic, 2025). A compelling approach for organizations to defend effectively is to simulate the future and forecast attacks before they occur. In this context, speed alone is insufficient; global firms must aim to anticipate attacks and act proactively, rather than reactively. This paper proposes a novel Darwin-Gödel machine cyber range to accurately forecast cyber attack before AI adversaries win, alongside a strategic go-to-market plan for a chosen cybersecurity organization, SimSpace, and associated risk mitigations.

World's First AI Cyber Attack at Scale

The world's first verified large-scale, AI-orchestrated cyber attack successfully compromised over 30 organizations globally in late 2025. Cyber threat actors, attributed to an elite Chinese state-sponsored espionage group, leveraged the popular code assistant Claude Code to carry out attacks with up to 90% autonomy, requiring human involvement only for initial access and data exfiltration (Anthropic, 2025). According to Anthropic (2025), this event demonstrates that “barriers to performing sophisticated cyberattacks have dropped substantially” and marks an “inflection point” where “AI models [have] become genuinely useful for cybersecurity operations, both for good and for ill.”

In this AI-fueled threat landscape, CISOs are increasingly challenged (Clark & McMillan, 2025). Human defenders cannot respond quickly enough to mitigate attacks, leaving organizations vulnerable to highly sophisticated, AI-driven adversarial campaigns (McMillan, 2025). In cybersecurity, attack dwell time refers to the interval between a successful intrusion and its detection by defenders. Even if dwell time approaches zero because cybersecurity professionals and technologies are getting better at detecting them, near-future attacks may achieve their objectives almost immediately, with faster breakout times than defensive responses can act (Castro, 2025). Without AI-centric defensive strategies, organizations face slow, inadequate, and costly outcomes as the pace and sophistication of AI-enabled attacks continue to accelerate.

Prediction as a Proxy for Power

Just as earlier generations struggled to grasp the implications of advances in arithmetic for the development of computing, contemporary society is still in the early stages of understanding and realizing the full significance of rapid advances in predictive modeling. The contributions of Vaswani et al. (2017) in “Attention Is All You Need,” which enabled the emergence of large language models, represents only an initial step in this broader transformation in what prediction really means for society.

Agüera y Arcas (2025) asserts that prediction *is* intelligence and that “everything alive is a computer.” By this logic, it is no longer sufficient to dismiss advances in AI as mere next-token prediction models that are prone to occasional hallucinations; in fact, this may be analogous to the predictive processes occurring in human brains. Classical definitions of intelligence emphasize the focus on reducing uncertainty for the decision-maker and managing information (Wheaton & Beerbower, 2006). However, framing true intelligence as prediction implies that information becomes obsolete the moment it is produced, whereas prediction is continuously updating to anticipate the next action.

The same can be said for rapid decision-making in an attack setting. In military aviation, the concept of the OODA loop (Observe, Orient, Decide, Act) guides rapid decision-making in trying to predict the adversary’s next move, and it has been adopted in cybersecurity to facilitate timely and effective defensive responses (Husák et al., 2022). The underlying principle is that the actor who completes the OODA loop most rapidly gains the advantage, a fact especially evident in high-speed aerial dogfights where survival depends on decision speed. The same principle

applies to cybersecurity: if AI-enabled attacks allow adversaries to spin the OODA loop faster than human defenders, innovation becomes not only necessary but existential. AI defenses must outpace AI advances to adequately defend.

Inspiration from Probabilistic Ensemble Analysis and Digital Twins

Promising research from other fields offers potential insights for addressing such cybersecurity challenges. For instance, Google GenCast employs weather prediction techniques based on probabilistic ensemble analysis to simulate future weather patterns with high accuracy (Price et al., 2023). Rather than producing a single forecast, the system iterates historical data across multiple simulations, up to fifty ensemble models simultaneously, allowing it to capture the range of likely outcomes. Price et al.'s (2023) advancements in operational weather forecasting suggest that analogous approaches could perhaps be applied to AI-driven, operational-level cybersecurity decision-making.

Similarly, NVIDIA Omniverse introduces innovations in physics-based simulation. By creating 3D digital twins of environments, such as cities, vehicles, or entire physical systems, users can evaluate the consequences of decisions before they affect the real world. Supporting the value of early intervention, Stecklein (2010) reported that for NASA, identifying errors during requirements or design phases is approximately 78x less expensive than addressing them in the testing phase and up to 1,500x less costly than correcting errors in production or operational environments. Anecdotally, product CISOs from large technology firms attending an Innovate Cybersecurity Summit (2025) emphasized that addressing coding errors and bug failures in pre-production cost roughly 1/40th of the expense incurred when such failures occur in production environments, where every second can directly affect consumer revenue.

This presents a fundamental opportunity for CISOs: How can one understand whether a cybersecurity technology or AI agent performs as claimed before deploying it in production environments? As industry expert Kevin Mandia (2016) observed, “attribution is a proxy for risk.” If knowledge of the adversary constitutes a form of understanding, then predictive insights may similarly serve as a proxy for actionable defense.

Digital Mirror & Adversary Emulation to Help Simulate the Future

Leading militaries, ministries of defense, and top technology firms are turning to digital twins and cyber ranges to train AI models, test AI agents, and validate agentic workflows in a similar manner to GenCast and Omniverse (Vellante & Gilbert, 2025). Cyber ranges function as

a type of training environment which leverages a digital representation of an organization's network. Unlike digital twins, which are exact replicas that can be large, costly, and difficult to maintain, cyber ranges often employ digital mirrors, approximations of reality that capture essential features while remaining sufficiently realistic.

Within this AI training environment, cyber ranges employ both user and attack emulation to produce a continuously evolving system shaped by reciprocal adaptation rather than a static sequence of scripted events (SimSpace, n.d.). Changes in user behavior influence the trajectory and effectiveness of attacks, while attack progression in turn alters user activity, creating an integrated feedback loop that extends beyond the isolated simulation of adversarial actions. Organizations such as SimSpace are developing adaptive attack generators that ingest threat intelligence, map adversary tactics, techniques, and procedures to libraries of atomic actions, and autonomously generate emulated campaigns. These campaigns evolve not only in response to shifts in the threat landscape but also in response to observed user behaviors within the environment. By executing such simulations at scale and frequency, cyber defenses can leverage AI-driven systems to co-evolve offensive and defensive behaviors, enabling training against both emerging attack patterns and the dynamic human responses that shape their real-world impact.

Evaluating a Darwin-Gödel Cyber Range for Preemptive Cyber Defense

One example of a self-reinforcing learning environment that operates on similar principles of learning from itself is the Darwin-Gödel machine (Zhang, Hu, Lu, Lange, & Clune, 2025). Gödel machines are named after mathematician Kurt Gödel and are inspired by his meta-learning concepts related to "learning to learn." They are co-named "Darwin" to reflect their incorporation of Darwinian evolutionary dynamics, particularly the capacity for internal self-improvement without reliance on external stimulus. In technological systems, the authors describe these machines as capable of modifying their own code and empirically validating those changes against industry-standard software benchmarks, achieving performance improvements of approximately 15 to 30 percent over human programmers (Zhang et al., 2025). To explore its utility, I differentiate, below, between reinforcement learning (RL), Gödel and Darwin Gödel variants.

Reinforcement learning models adaptive behavior as the problem of maximizing expected cumulative reward through interaction with an environment. An agent learns by updating a parameterized policy or value function using experience-driven feedback, commonly

via methods such as temporal-difference learning or policy gradient optimization. While the agent's parameters evolve over time, the learning algorithm itself remains fixed: the update rules, optimization objective, and representational assumptions are specified a priori and do not change during learning. Consequently, reinforcement learning optimizes behavior within a predefined learning framework and relies on empirical trial-and-error rather than formal guarantees of global optimality. This architectural separation between a static learning rule and dynamic policy parameters limits the degree of self-adaptation achievable in standard RL systems.

Gödel machines generalize adaptive learning by permitting self-modification of the entire system, including the learning algorithm itself. A Gödel machine is a self-referential computational system equipped with an explicit utility function and a formal proof system capable of reasoning about its own code. The system systematically searches for proofs that executing a particular self-rewrite will increase expected utility relative to continued execution without modification. Upon finding such a proof, the machine commits to the corresponding self-change. This framework yields strong theoretical guarantees of optimality relative to the assumed axioms and utility function, as any executed modification is provably beneficial. However, the reliance on formal proof search over rich computational hypotheses makes Gödel machines computationally infeasible in realistic environments, confining their role primarily to theoretical analyses of optimal self-improving agents.

Darwin Gödel machines relax the proof constraints of Gödel machines by replacing deductive justification with evolutionary selection over self-modifying programs. In this framework, multiple variants of an agent are generated through stochastic self-modifications, potentially affecting learning rules, internal representations, or decision procedures. These variants are evaluated empirically based on observed performance, and selection favors those that achieve higher utility. While formal reasoning may still be used to guide or restrict the space of allowable modifications, improvement is ultimately determined through empirical competition rather than formal proof. This approach abandons the global optimality guarantees of Gödel machines in favor of practical tractability, enabling adaptive modification of learning processes under realistic computational constraints and positioning Darwin Gödel machines between conventional reinforcement learning and fully proof-based self-improving systems.

As CISOs seek to defend their organizations more efficiently and evaluate technologies that may outperform human analysts, applying the Darwin–Gödel machine concept to a cyber

range environment for testing AI-driven cybersecurity tools represents a cutting-edge innovation.¹ These systems use machine-learning-based decision frameworks to emulate human defensive behavior in responding to cyber threats. New AI startups seek to meet or exceed human performance because of the imminent future in which AI attacks will be faster than humans can respond (Clark & McMillan, 2025). Leading cybersecurity vendors are now expanding research and development into autonomous multi-agent frameworks that aim to augment and in some cases supersede traditional human-centric operations (Torq, 2024).

Whether for leading cybersecurity vendors developing autonomous Security Operations Center agents or Global 2000 enterprises seeking stronger defensive readiness, the concept of a Darwin–Gödel Cyber Range functions as a pre-production training and testing environment for AI. In this setting, the range generates non-personally identifiable synthetic data, feeds logs into security technologies such as Endpoint Detection and Response (EDR) and Security Information and Event Management (SIEM) platforms, exports that correctly labelled data into notebook environments like Google Colab through pipelines such as BindPlane, and enables the system to iteratively learn from its own simulated activity thousands of times per day. Following standard machine learning recommendations, models will be trained on approximately 80 percent of the available data and evaluated on the remaining 20 percent as a test set to estimate performance (Kohavi, 1995). To further reduce overfitting risk and ensure continuity, performance will also be assessed through external or out-of-sample validation using data not involved in model development, which will simply be generated from a separate isolated cyber range tenant (Hastie, Tibshirani, & Friedman, 2009). The following updated Double Diamond model depicts a practical workflow for this proposal:

¹ To avoid conflating ‘AI’ with popular generative large language models, this paper focuses specifically on the development of autonomous Security Operations Center (SOC) agents designed to identify malicious activity on a computer network, although the proposed framework could likely be used to evaluate generic agents of any kind.

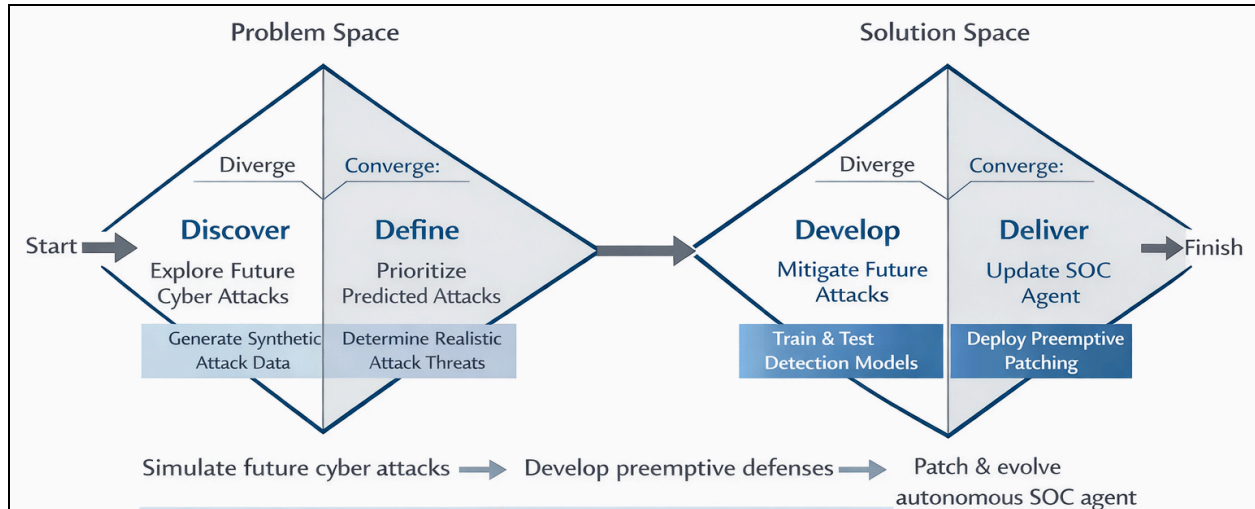


Figure 1: Double Diamond model adapted to the Darwin-Gödel Cyber Range Proposition

For this model to continuously adapt to itself and to emerging cyber threats, it must engage in ongoing cycles of unlearning and relearning. Rather than relying on external human interventions, the predictive model employs a reinforcement learning algorithm in conjunction with newly generated synthetic data to distribute errors² and updates. This approach allows the system to recommend defenses that improve upon existing policies while estimating discrepancies between actual outcomes and forecasted updates (Candidate 1097682, 2025). Even if the observed relationships between attack emulations and recommended defenses are correlative rather than causative, the model can still generate meaningful evasions for preemptive cybersecurity applications, recently referred to as Automated Moving Target Defense (AMTD) in pioneering projects.

Beyond its general applicability to a range of cybersecurity challenges, Gartner (2023) projects that by 2032, at least 30 percent of all Attack Surface Management (ASM) solutions will incorporate Automated Moving Target Defense (AMTD) techniques. More recently, Gartner (2025) notes that management of cybersecurity exposure is evolving into a preemptive discipline that leverages intelligent simulation, AI-enabled remediation, and automated defenses.

Mitigating Risks

Building on the foundation of Drenth's (2025) Firm Strategy for AI, I will also apply Drenth's (2025) TUNA model³ to evaluate risk mitigations and the dangerous invisible status quo, often referred to as the "ghost scenario."

² such as forward-pass retraining or backpropagation

³ Turbulence, Unpredictability/Uncertainty, Novelty, and Ambiguity

TUNA Model Evaluation

Turbulence. Turbulence is a critical risk to evaluate because disruptions continually reshape economic incentives in a crowded market⁴, and SimSpace could easily overextend, raise too much capital, or scale too quickly. To assess a more disciplined and resilient growth approach, I apply Trevor’s Spatial Differentiation model for evaluating organizational Size⁵, Market scope, and Product variety (Trevor, 2025).

Three scenarios are plotted: the number of markets targeted on the X-axis, the size of the global go-to-market team on the Y-axis, and the number of new AI products introduced on the Z-axis.

Scenario 1. (Current, purple): 3 people, 3 markets, 3 AI products → 27 volume units.

Scenario 2. (Future, red): 7 people, 3 markets, 6 AI products → 126 volume units (~4.7x more complex than current state).

Scenario 3. (Proposed, green): 7 people, 3 markets, 3 AI products → 63 volume units (only ~2.3x more complex than current state), offering greater scalability while limiting the risk of confusion, turbulence, and overextension. This option is best fit for navigating turbulence while scaling responsibly.

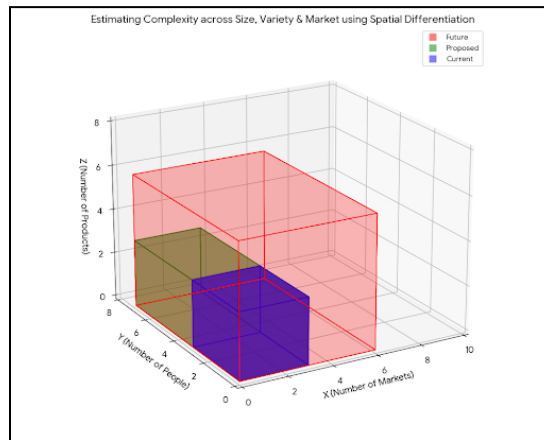


Figure 3: Estimating Complexity of Size, Variety & Market Using Trevor’s (2025) Spatial Differentiation

The guiding principle, here, is to intentionally constrain surface area and complexity so that alignment, execution quality, and organizational resilience remain strong as SimSpace scales.

⁴ Sometimes referred to in startup cultures as a ‘red ocean’

⁵ Referred to as SVM Matrix

Unpredictability and Uncertainty. George Box is widely cited saying that “all models are wrong, but some are useful,” implying there are inherent limitations of any framework in fully capturing real-world complexity (Box & Draper, 1987). This limitation is especially important for our AI training environment that adapts and learns from itself, where emergent technological leaps introduce additional uncertainty. Given this context, safety and risk governance will be treated as critical factors. SimSpace aims to incorporate an AI safety architecture within the cyber range to constrain unintended behaviors and reduce uncertainty during go-to-market execution for the proposed Darwin–Gödel learning environment. In practice, this approach includes engaging domain experts with experience designing reinforcement-learning ecosystems and safety-bounded autonomous systems, thereby strengthening oversight, guiding system evolution, and reducing exposure to unforeseen outcomes.

Novelty. Assessing the influence of novelty within our go-to-market strategy highlights the risk of emerging geopolitical developments or new technological innovations disrupting execution and slowing strategic momentum. SimSpace works closely with the Japanese Ministry of Defense to strengthen national cybersecurity readiness, and recent policy developments illustrate how rapidly conditions can evolve. Japan’s enactment of the Active Cyber Defense Law grants authorities – including the National Cybersecurity Office, National Police Agency, and Japanese Self-Defense Forces – the ability to preempt cyber threats under defined risk thresholds (Dobell & O’Neill, 2025). In parallel, Prime Minister Takaichi-San recently indicated that Japan would support Taiwan in the event of an attack by the People’s Republic of China, also in context of continued PRC incursions into Japan’s Senkaku Islands (Reuters, 2025). Novel geopolitical triggers may increase the likelihood that cyber operations could go ‘kinetic,’ shifting from the digital to the physical domain (NHK World–Japan, 2026). In this environment, our strategy emphasizes mitigation of brand and partner-risk exposure by aligning our offerings with international law, national cyber-defense regulations such as ISMAP⁶, and emerging AI governance frameworks.

Ambiguity. Drenth (2025) describes the importance of assessing a “ghost scenario” in strategic planning, which refers to an implicit course of action that carries real organizational consequences yet remains largely unseen or undefined. This concept closely aligns with

⁶ Information System Security Management and Assessment Program (ISMAP)

ambiguity, in which strategic intent or execution pathways are difficult to interpret or predict. To mitigate ambiguity in our go-to-market approach, we explicitly expose multiple strategic scenarios, including the current default trajectory. In this ‘ghost’ scenario, our existing non-AI product portfolio expands with incremental features that enable AI capabilities, internal teams successfully upskill, and customers readily adopt the new functionality without disruption. However, recognizing that this outlook is too optimistic, an alternative and more conservative pathway is being considered for global markets. This differentiated approach narrows delivery scope, phases Darwin-Gödel capabilities gradually toward year-end, and incorporates regional nuances such as language localization and alignment with allied government mandates.

Ambiguity is further managed through quarterly senior leadership reviews to reassess progress and, consistent with the Lean Startup framework, determine whether to pivot or persevere (Ries, 2011).

Conclusion

CISOs must reimagine their security posture as their AI-fueled threat landscape continues to evolve, putting less emphasis on upskilling human operators and more emphasis on developing capabilities to rapidly respond or even preempt cyber attack. As adversaries increasingly employ autonomous and self-improving AI systems to discover and exploit vulnerabilities at machine speed, cybersecurity leaders must cultivate defensive environments that learn, adapt, and respond with equal sophistication.

This discussion argued that a Darwin-Gödel cyber range represents a cutting-edge innovation and promising paradigm for advancing such capabilities by enabling organizations to simulate future attack trajectories, generate synthetic data, and derive actionable defensive insights prior to real-world compromise. When coupled with a disciplined go-to-market strategy and robust risk mitigation mechanisms, this proposal offers a viable pathway for operationalizing preemptive cyber defense at scale. The broader implication to the industry and to the allied world is that resilience in the emerging era of AI-enabled cyber defense will depend not only on faster reaction, but on a Government or organization’s capacity to forecast, experiment, and evolve in advance of cyber attack. By simulating the future in this manner, CISOs will not just be able to act faster, they will act first.

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